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Study-branch: MSc. Artificial Intelligence (120 ECTS)

Advanced Statistics Workbook (DLMDSAS01)

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I Abstract

This workbook documents the comprehensive application of core concepts in *Advanced Statistics (DLMDSAS01)*, covering probability distributions, parameter estimation, hypothesis testing, and Bayesian statistics, utilizing a personalized dataset specified by the ξ parameters.
The initial tasks involved fundamental analysis of probability distributions and descriptive visualization:

- **Task 1** analyzed basic probabilities and visualizations. Since $\xi_1 = 3$, the task involved modeling the number of meteorites falling into an ocean using a *geometric distribution* counting the number of Bernoulli trials until success, with $p = \xi_2 = 0.61$. This task required calculating the probability mass function (PMF) up to the point where the probability drops below 0.5%, calculating the *expectation* and *median*, and displaying these results graphically.

- **Task 2** derived the *Probability Density Function (PDF)*, computed the waiting time probability, and analyzed the location and dispersion parameters (mean, variance, and quartiles) for a continuous waiting time variable (Y) modeled by a mixture of exponential functions. This task included visualizing the PDF and a discrete histogram.

- **Task 3** applied the theory of *Transformed Random Variables* to derive the Gamma distribution of the total failure bandwidth ($T = S_1 + S_2$) for a dual-router system, based on the assumption of independent and identically distributed (i.i.d.) router failures. The model parameter (θ) was estimated using *Maximum Likelihood Estimation (MLE)* by maximizing the derived log-likelihood function.

The subsequent tasks demonstrated advanced inferential and predictive modeling techniques:

- **Task 4** performed a *Hypothesis Test* to assess whether a new factory system produced significantly higher hammer weights ($\mu_{new} > 904$ g), as mandated by $\xi_{13} = 2$. This involved proposing null (H_0) and alternative (H_1) hypotheses, constructing the Z-score test statistic, defining a decision rule based on critical values, and calculating error probabilities.

- **Task 5** tackled model complexity and overfitting through *Regularized Regression*, fitting a *10th-degree polynomial* (as $\xi_{15} = 2$) to highly variable sample data (ξ_{16}). Solutions derived using Ordinary Least Squares (OLS) were explicitly contrasted with the shrinkage effect achieved via *Ridge regularization* (L2 penalty), emphasizing the trade-off between bias and variance.

- **Task 6** focused on *Bayesian Estimation*, determining the posterior distribution of the scale parameter (θ) for a Gamma distributed variable. Utilizing the property of *conjugate priors* (Inverse Gamma family), the posterior distribution was derived and used to calculate Bayes point estimates associated with both the square-error loss function (mean) and the mode of the posterior distribution.

The methodology throughout employed rigorous mathematical derivation and utilized Python tools for computation, estimation, and comprehensive *data visualization*. This report successfully demonstrated proficiency in applying and interpreting both frequentist and Bayesian approaches to statistical inference.

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IV Abbreviations

AFL	American Fuzzy Lop
API	Application Programming Interface
BIOS	Basic Input/Output System
Brick	Binary Run-time Integer Based Vulnerability Checker
CaaS	Container as a Service
CAB	Change Advisory Board
CE	Community Edition
CI	Continuous Integration
CLI	Command Line Interface
CNCF	Cloud Native Computing Foundation
CRED	C Range Error Detector
Dev	Development, the development team

1 Latex

1.1 Tools

MiKTeX: https://miktex.org/download TeXLive: http://tug.org/texlive/ (or alternative LaTeX-systems).

A good editor is essential. Sometimes combined editors and compilers (e.g. TeXShop) can be really productive. Make sure you learn the use of synchronized navigation then.

A vector graphic is one where strokes remain strokes even at the highest resolutions: e.g. the Figure **??** or the Logo on the Titelblatt (notice: you can click from here to there). Many tools generate vector-graphics for plots from any data-set. E.g. Plotly (with the use of the Browser-Print), MatPlotLib or even OpenOffice, LibreOffice or MS-Excel.

1.2 Literature References

1.3 Pictures

Task 1: Basic Probabilities and Visualizations (1)

$\xi_1 = 3$

$\xi_2 = 0.61$

A geometric distribution counting the number of Bernoulli trials with p $ \xi_{2} $ until it succeeds.

```
#import libraries
import numpy as np
import matplotlib.pyplot as plt
from scipy.stats import geom
```

Step 1:

Define Model

Define $X \sim \text{Geometric}(p)$

and state the Probability Mass Function (PMF): $P(X=k)=(1-\xi_2)^{k-1}\xi_2$.

```
# Define parameters based on xi_2
prob_success=0.61
prob_faliure=1-prob_success
# Initialize the Geometric distribution object (k >= 1)
dist = geom(prob_success)
```

Step 2:

Compute $E[X]$, using the formula $E[X]=1/p$ for $k \geq 1$ (Unit: 2.1, P. 48)

$$E[X]=\frac{1}{0.61} \approx 1.6393$$

```
E_X = 1 / prob_success
print(f"E[X]: {E_X:.4f}")
```

```
E[X]: 1.6393
```

Step 3:

Compute the median M: find the smallest integer k such that the CDF satisfies $P(X \leq k) \geq 0.5$. Use $P(X \leq k)=1-(1-p)^k$.

$ *_{(Unit 2: 2.1, P. 50)}$

We solve for the smallest integer k such that:

$$(1-0.61)^k \leq 0.5$$

$$(0.39)^k \leq 0.5$$

$$k \geq \frac{\ln(0.5)}{\ln(0.39)} \approx 0.736$$

The smallest integer satisfying this is $k=1$.

```
# The median is found using the inverse CDF (Percent Point Function,
ppf)
M = dist.ppf(0.5)
print(f"Median M = {M:.0f}")
```

```
Median M = 1
```

Step 4:

Determine the upper limit K by finding the smallest integer k for which $P(X=k)<0.005$. Explicitly cite the proof steps.

```
threshold = 0.005
k_values = []
# Iterate k starting from 1
for k in range(1, 15):
    prob = dist.pmf(k)
    print(f"P(X={k}) = {prob:.6f}, P(X=k) = {(prob_faliure**(k-
1))*(prob_success)}  Condition P(X=k) < 0.005$? {prob < threshold}")
    if prob < threshold:
        # The range ends at k-1, since k itself is below the
threshold.
        break
    k_values.append(k)

probabilities = dist.pmf(k_values)
K_max = max(k_values)
print(f"The visualization range is k = 1 to {K_max}")
```

```
P(X=1) = 0.610000, P(X=k) = 0.61 Condition P(X=k) < 0.005$? False
P(X=2) = 0.237900, P(X=k) = 0.2379    Condition P(X=k) < 0.005$? False
P(X=3) = 0.092781, P(X=k) = 0.092781  Condition P(X=k) < 0.005$? False
P(X=4) = 0.036185, P(X=k) = 0.03618459Condition P(X=k) < 0.005$? False
P(X=5) = 0.014112, P(X=k) = 0.014111990100000003 Condition P(X=k) <
0.005$? False
P(X=6) = 0.005504, P(X=k) = 0.005503676139000001 Condition P(X=k) <
0.005$? False
P(X=7) = 0.002146, P(X=k) = 0.0021464336942100004     Condition P(X=k)
< 0.005$? True
The visualization range is k = 1 to 6
```

Step 5:

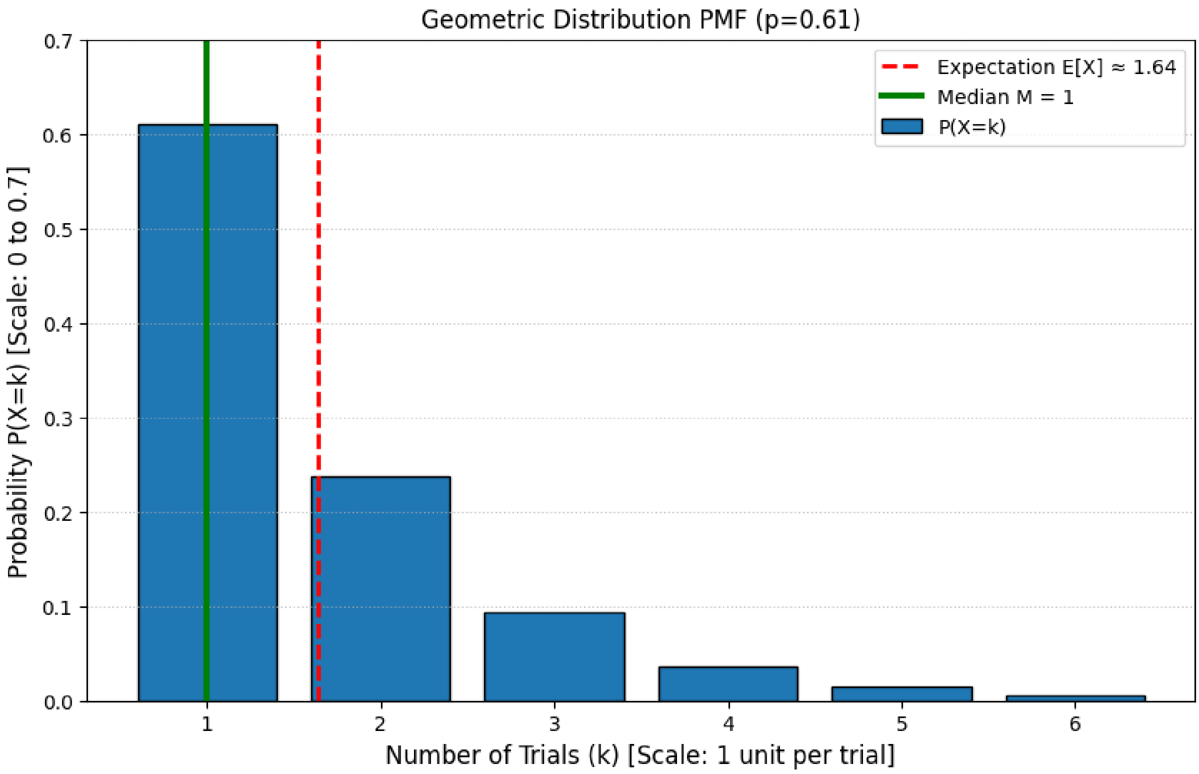
Generate the graphic (bar plot/discrete histogram) for $k=1$ up to K. Include the scale. Present the calculated $E[X]$ and M within the graphic.

We generate a bar plot to visualize the discrete probability mass function (PMF)

$ *_{(Unit 5: 5.5, P. 139)}$

The graphic must include the scale of the axes and clearly display the calculated Expectation and Median

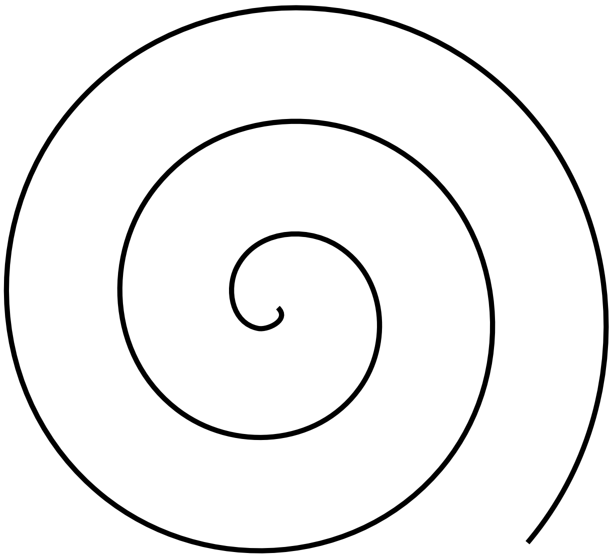
```
plt.figure(figsize=(10, 6))

# Plotting the PMF using a bar plot
plt.bar(k_values, probabilities, color='#1f77b4', edgecolor='black',
label='P(X=k)')

# Mark Expectation
plt.axvline(E_X, color='red', linestyle='--', linewidth=2,
            label=f'Expectation E[X] ≈ {E_X:.2f}')

# Mark Median
plt.axvline(M, color='green', linestyle='-', linewidth=3,
            label=f'Median M = {M:.0f}')

# Formatting and Scale (required by assignment)
plt.title(f'Geometric Distribution PMF (p={prob_success})')
plt.xlabel('Number of Trials (k) [Scale: 1 unit per trial]',
fontsize=12)
plt.ylabel('Probability P(X=k) [Scale: 0 to 0.7]', fontsize=12)
plt.xticks(k_values)
plt.ylim(0, 0.7) # Explicitly setting Y scale for clarity
plt.grid(axis='y', linestyle=':', alpha=0.7)
plt.legend()
plt.show()
```



Justification of Tool Usage:

We utilize Python's standard statistical (scipy) and plotting (matplotlib) libraries.

These tools are widely accepted and trusted for producing accurate numerical results and visualizations in scientific computation.



1.5 Tables

“ Industrial era ”	“Jobs ”	“ Wanted: Upgrade”’
Parts exchanger	Fitter	mecatronics special-ist
eShop	reseller	“ Client-suggester”
“ Coding-guru”’	Softwaredesign	Whole-life designer
JA! Gut & Günstig	brand-names	“ Life-Style Feeling”
Internetbanking	Bank clerk	Customer adviser
Robots	Specialist	Machine supervisor
Bush	Gardener	Nature-sculptor
Painting	Painter	Interior Design

1.6 Listes

- one
- twoi
- threei

1. first
2. second
3. third

1.7 Formulæ

A formula can be written inline, e.g. as $\frac{d}{dx}\mathrm{arctg}(x) = \frac{1}{1+x^2}$ or, in centered math:

$$\frac{d}{dx}\mathrm{arctg}(x) = \frac{1}{1+x^2} \tag{1.1}$$

Notice that formulæ that are centered start bigger (technically, they start in `\displaystyle`) than they start inline (technically, they start in `\textstyle` all subsequents reductions, e.g. an exponent, goes to

`\scriptstyle` then `\scriptscriptstyle`). Indeed a best effort is made so that inline formulæ do not change the line height which would bother the eye of a reader.

Formulæ can be given a number and a label. Numbering happens automatically with `\begin{equation}` and `\end{equation}` and can be avoided if enclosing the formula between `\[` and `\]`. If using the `\label` macro inside, you can refer automatically to this equation using `\ref{label}`. E.g. Thanks to equation 1.1 one dare say that:

$$\int_0^t \frac{1}{1+x^2} dx = \arctan(t) \tag{1.2}$$

1.8 Tools and Code

Many users of this template will want to include some code.

The simplest way to do so is to use the `\verb` macro which is followed by a sign, then some code, then the sign again to close. This is the inline version which works as in:

which yields:

The multiline version of this is called `\begin{verbatim}` and finishes with `\end{verbatim}`.

1.9 Citation examples

Monography (*Advanced Statistics DLMDSAS01*, 2025, p. 22)

1.10 Task 1

Bibliography

Advanced Statistics DLMDSAS01. (2025). IU Internationale Hochschule GmbH, IU International University of Applied Sciences.